Milestone 1 (20th March, 5pm), 10% of the final grade

Dataset

The selected dataset is named "World Athletics Database". It contains csv files in total detailing 461 523 athlete performances for various events scraped from the website world athletics, which shares performances and world records for both men and women athletes in various track and field categories, such as 100m women. Amongst the 99 csv files available, 48 relate about feminine events, 49 for male events and 2 extra CSVs compiling the data from all events, regardless of the gender, and the current world records.

For convenience, only the two extra csv files will be used as they contain all of the data compiled for each event and the world records, which are the necessary information needed for our study.

Description

The data gathers athlete performances at different events all around the world. More specifically the the name of the competitor, his/her gender, date of birth and nationality but also the event he/she is competing in, the date of the event, the position achieved, the wind (expressed in meters per seconds), 2 columns mark displaying the results in two different formats (durations in seconds or in format hours:minutes,seconds), and a last one containing the "Results Score".

Problematic

Our website targets sports fans or anyone interested in human sportive performance. We aim to create a playful visualization that allows users to explore the evolution of athletic achievements and discover patterns behind world records.

Through our visualization, we aim to understand how elite athletic performance evolves over time and across discipline, age, gender and nationality. By analyzing performance data across most disciplines of track and field, our visualizations aim to highlight patterns in performance progression, athlete characteristics, and the context in which records are achieved. By doing so, we would like to highlight improvements in technology, training and competition while also showcasing exceptional athletes.

Concretely, we would like to display athletic data in an interactive and playful way. Starting by making the user "choose an athlete" based on a selected gender,nationality, year of birth and event category. The athlete would then compete against his/her peers (absolute world record, his/her old self, Swiss record) to compare their performances.

Otherwise, visualizations will focus on several aspects such as:

- the temporal evolution of world records in different athletic competitions
- the profile of world record athletes (age, nation…) and whether some patterns can be detected
- the characteristics of athletes who (have) withhold(eld) world records in several events
- whether an athlete performs better when competing in its home country or not
- the "peak" age for athletes depending on their event

The visualizations aim to show that world records are not isolated achievements but integrated in larger patterns. They highlight performance improvement over time, which athletes have had the greatest impact on their sport, and when athletes tend to reach their peak performance.

Exploratory Data Analysis and pre-processing of the data set you chose

For the pre-processing of the data, we decided not to include relay events, as there were no athlete names provided, making the analysis difficult. We also chose to work exclusively with the data.csv file, as it was easier to handle.

Regarding data quality, we identified several missing values: 6,430 for the Date of Birth, 2,731 for the final position, and 336,236 for the wind variable, while the remaining columns contained no missing data. For each event, performance is recorded either in meters or seconds; to distinguish between these two categories, we introduced a binary flag variable equal to 1 when the result is measured in meters.

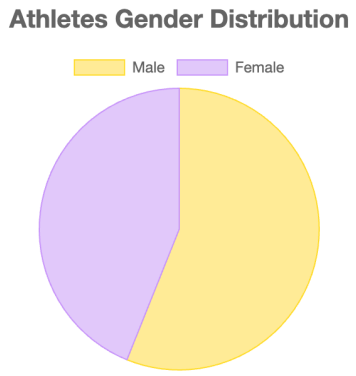
We also addressed redundancy by removing one of the duplicate “Marks” columns, retaining the version that reports times in seconds and distances in meters. When examining the “Age” column more closely, we found implausible values such as “-14” and “5.” Since we were unable to verify the corresponding dates of birth, we treated these entries as missing values.

Additionally, because the methodology used to compute the “results score” was unclear, preventing precise interpretation, we chose to drop this column. The remaining variables—gender, position, nationality, venue, and event date—were deemed reliable and suitable for analysis.

We also split the “Venue” field into two separate columns, “Place” and “Country,” to enable future analysis at the country level rather than only by venue.

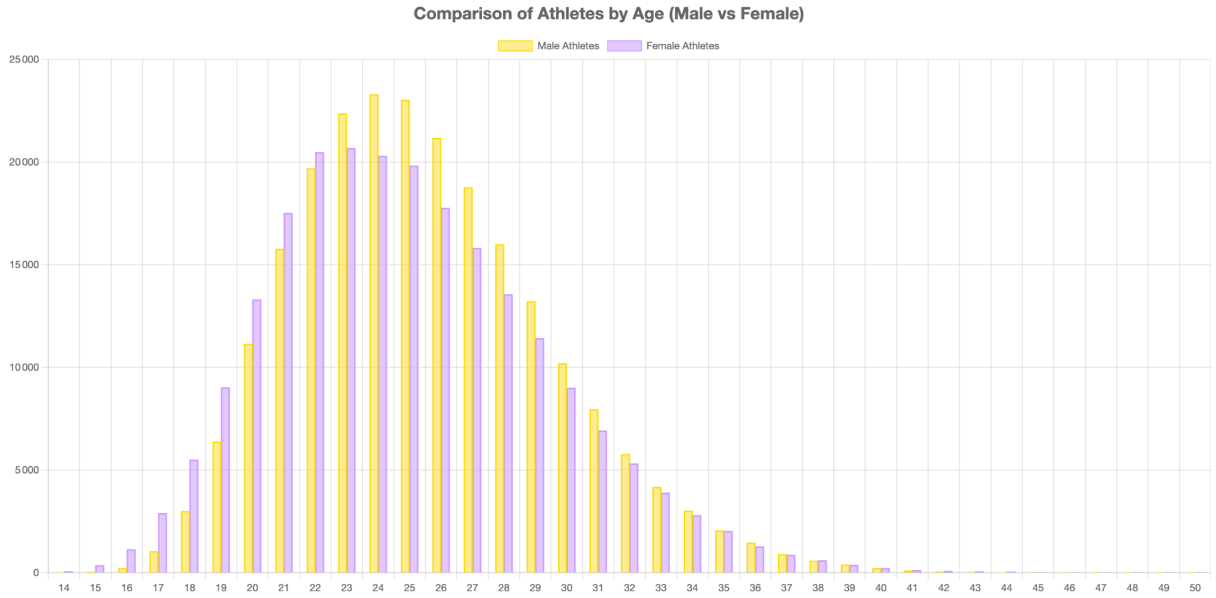
Aside from these decisions, the dataset was already clean and well-structured, so no further modifications were required. The final dataset was saved as final_data.csv.

For the exploratory data analysis:



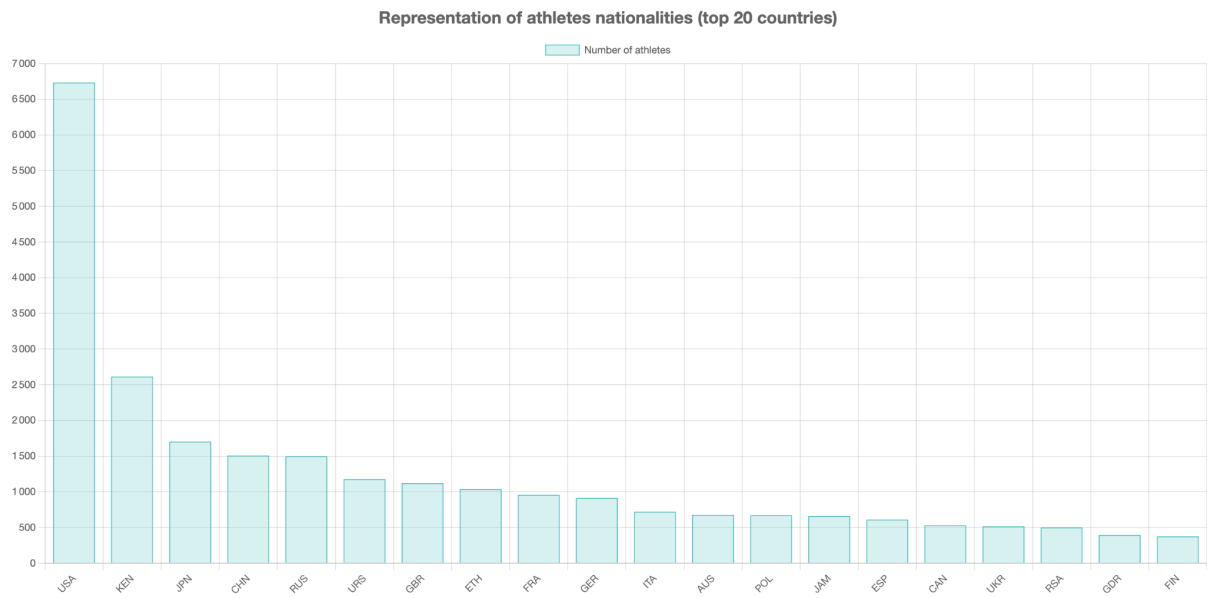
Female - 14 943
Male - 19 093

The gender distribution is relatively balanced, with 56.1% (19 093) of male athletes and 43.9% (14 943) female athletes in the dataset.



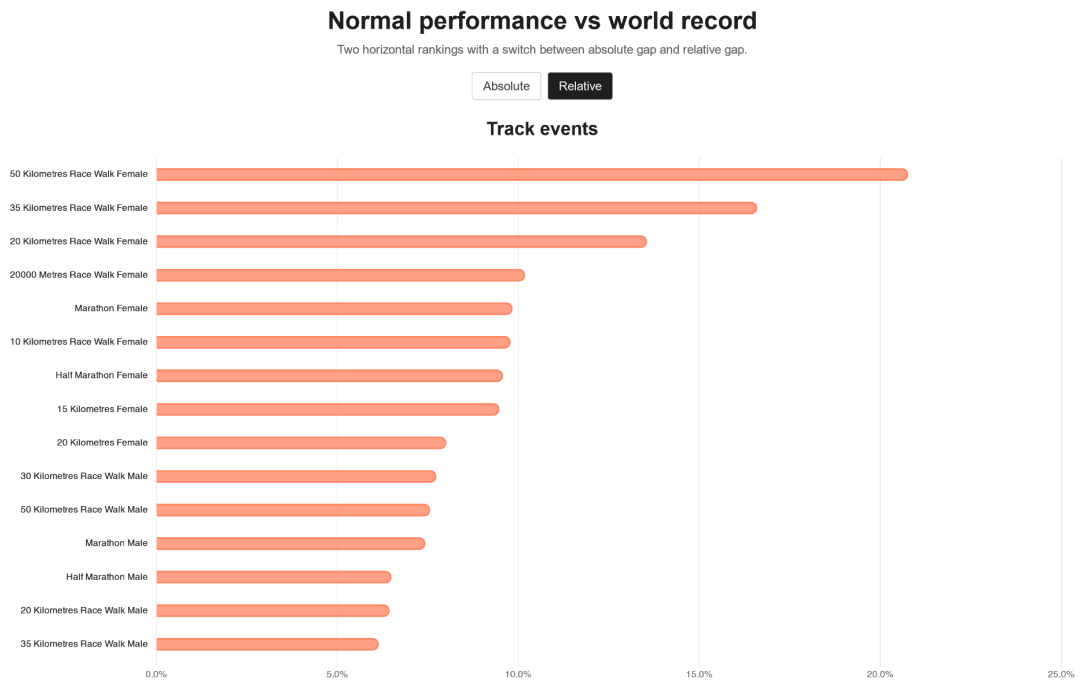
Overall, female athletes tend to be more represented at younger ages than male athletes. As illustrated in the graph, the distribution for women peaks earlier, around ages 22 to 24 (around 20 500 women for each of these age categories), whereas for men it peaks slightly later, around ages 24 to 25 (between 23 000 and 23 300).

Additionally, although men are more numerous overall, women are clearly more represented in the younger age groups (≤ 22), with for example 58% more women than men for the age of 19. This difference gradually decreases with age, as the male distribution becomes more dominant in the older age categories.

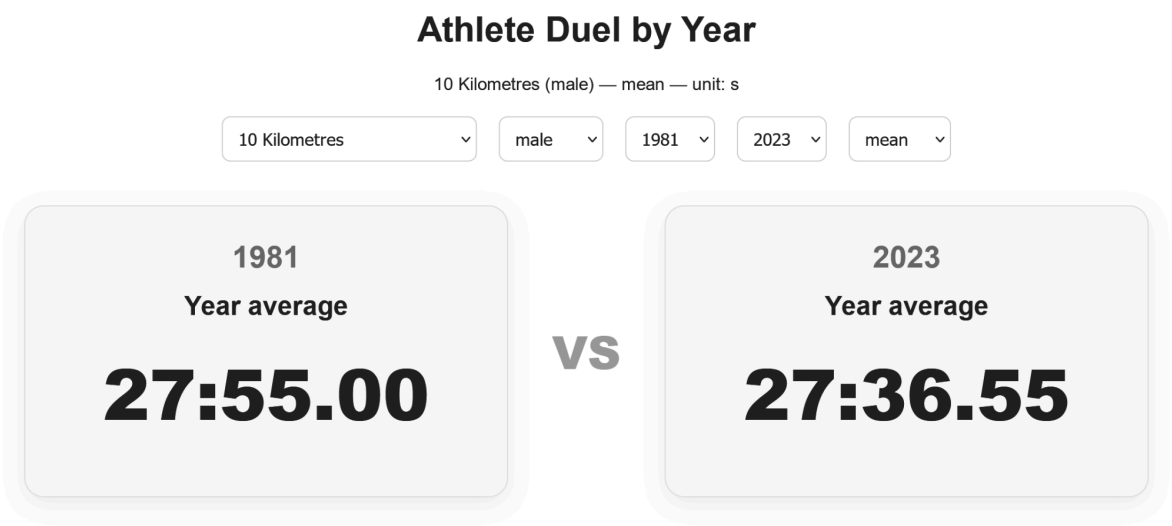


The graph shows the distribution of athletes' nationalities among the top 20 most represented countries. Overall, the data highlights a strong dominance of the United States with ~6 733 athletes , while participation among other countries is more evenly spread across the top 20.

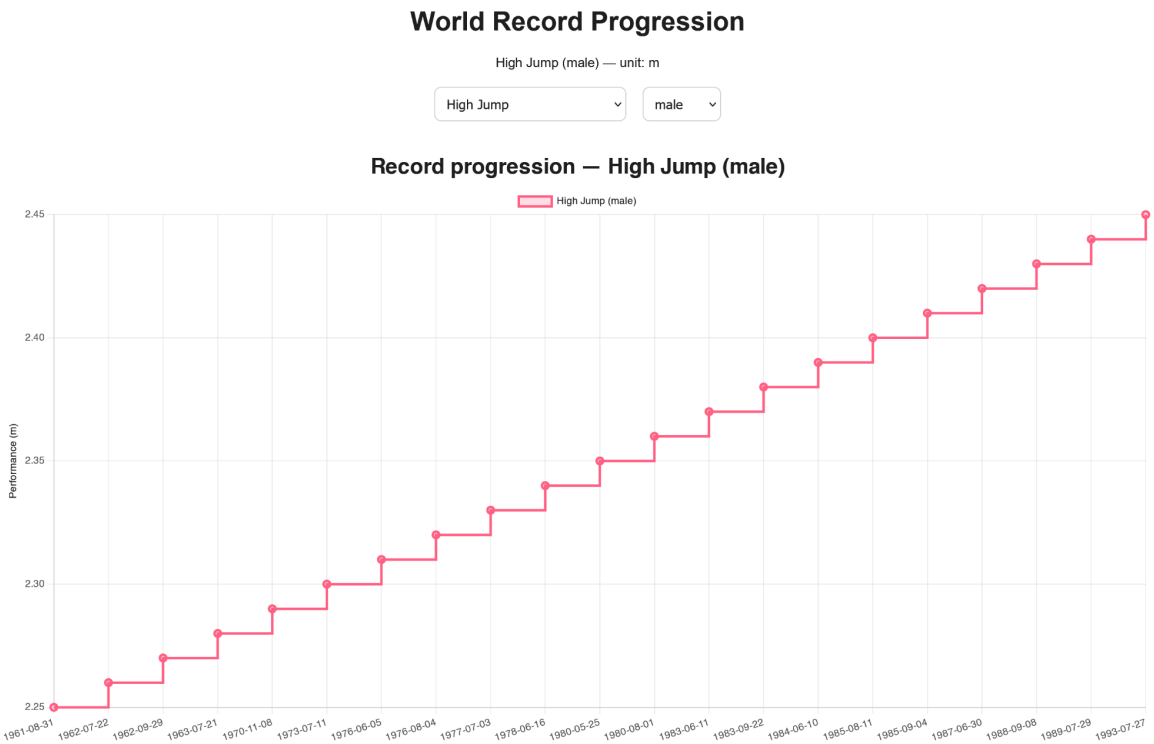
A second group of countries, including Kenya, Japan, China, and Russia, follows with moderately high representation (1 500 to 2 600 participants), though still far behind the United States. The remaining countries display gradually a lower number of athletes (ex: 500 for Ukraine).



This graph shows the difference between the world record and the mean of all the other results. We use relative because absolute has no meaning when we compare 100m with 50km for example. It highlights strong differences across disciplines particularly in women's events, has the largest relative distance between the world record and the rest of the results, while men's road-running events appear much denser near the record.



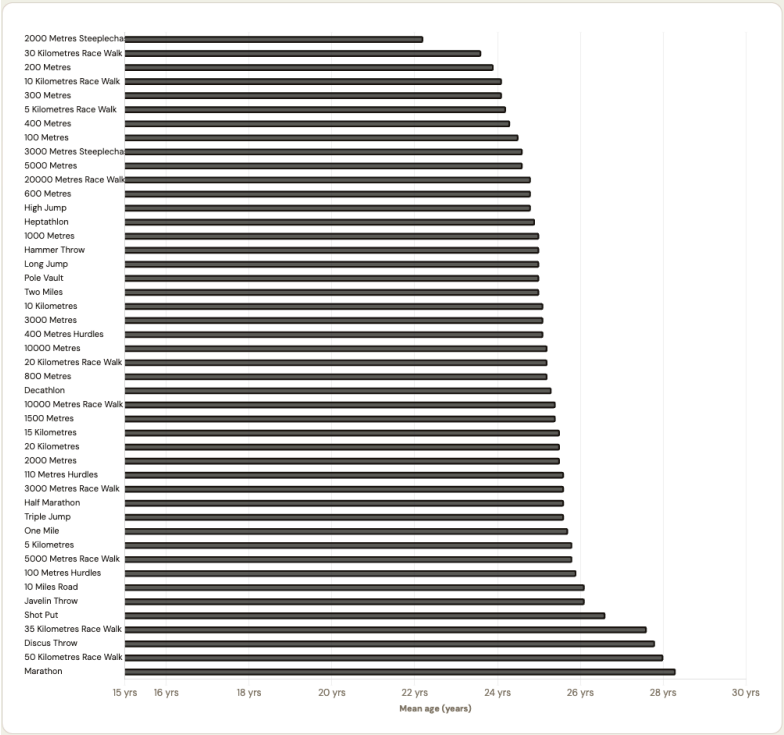
This graph is an interactive year-by-year comparison tool. It lets us choose an event, a gender, two years, and the statistic we want to compare, such as the mean result. In this example, we selected the men's 10 kilometres and compared the yearly average in 1981 and 2023.
It shows a small improvement in the yearly average between 1981 and 2023. More broadly, the graph helps highlight how the evolution of performance differs across events and periods.



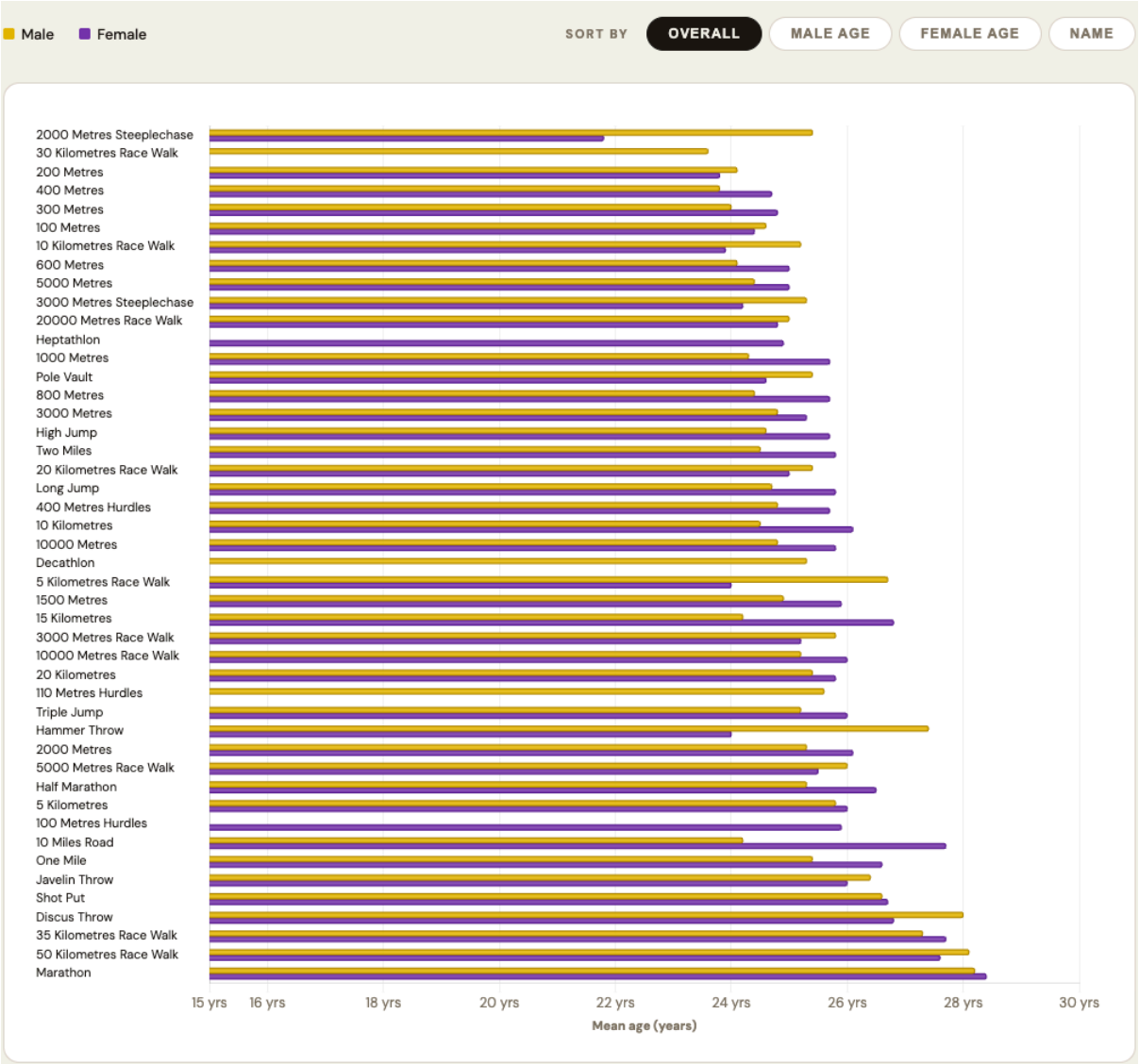
This graph shows the evolution of the world record over time for a selected event and gender. Here, we took the men's high jump as an example. The staircase shape reflects the fact that a record remains the same until someone breaks it. In this case, the record

progresses gradually from about 2.25 m to 2.45 m. By hovering over a point, we can also see who set the record and the exact performance.

We also wanted to explore the distribution of ages in a particular sport. This graph shows the overall distribution, and by hovering over a specific line, it is also possible to see the exact average.



Additionally, we analyzed the distribution by gender, as shown in the second graph. Some sports include only men, while others include only women. It was particularly interesting to observe the gaps between men and women in certain sports, as well as those where the situation is completely reversed. As in the other graph, it’s also possible to hover over a specific line and see the average for each gender.



Related work

Several analyses have already been conducted on Olympic data. We have found lots of different examples, like the Olympic History Data – A Thorough Analysis (https://www.kaggle.com/code/heesoo37/olympic-history-data-a-thorough-analysis) there they provide an exploration of the Olympic Games, covering multiple sports, geographic representation, height and weight of athletes and they even talk about nazis crushing the 1936 Art Competitions at Olympics.

We have also found datasets as this one more specialized datasets like Olympic Track & Field Results (https://www.kaggle.com/datasets/jayrav13/olympic-track-field-results/data) which focuses specifically on track and field events, offering detailed performance records for athletics competitions. However, we did not find any existing visualizations or analyses based on the exact dataset we used.

Although there is already a lot of work on the Olympics and sports data in general, fewer analyses focus specifically on athletics. Our project aims to take a closer look at this discipline.

Athletics is also a very diverse field, with many different types of events such as sprints or long-distance races over various distances, jumps, pole vault, heptathlon, etc.), which offers a wide range of possibilities for data analysis. We also wanted to present the data in a more engaging and interactive way, making it easier and more enjoyable to explore.

For our sources of inspiration, we explored the Oculi Mundi (https://oculi-mundi.com/) , which brings together a collection of antique maps. We especially liked the way it creates a narrative, giving the feeling of following a story as you browse.

We were also inspired by a Utopia Tokyo (https://www.utopiatokyo.com/) that combines Japanese-style masks with an imagined futuristic world. Its playful and immersive design makes you want to stay longer and continue exploring the different masks and find your own.

We also looked at Beyond Design Into Experience (https://15th.plus-ex.com/). We liked its strong storytelling approach and the way the whole experience unfolds through scrolling, motion, and transitions rather than through a traditional page structure. Its minimal black-and-white visual identity also shows how a website can remain visually striking while keeping a very clear and controlled design language.

Another source of inspiration was the Qubex Experience (https://qubexai.webflow.io/). What stood out to us was its energetic pacing and the way it mixes bold typography, animated sections, and interactive elements to guide the user through the content. Even though it is a product website, it feels more like a dynamic digital experience than a standard corporate page. We found this interesting because it shows how interaction design and visual rhythm can make technical information more engaging and easier to explore.